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Dwarf Seahorse (*Hippocampus zosterae*) Density, Distribution, and Habitat Use in Texas

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Abstract

Seagrass beds are composed of foundation species, providing essential nursery grounds, feeding areas, and refuge for various marine life. Several species of fish and invertebrates utilize seagrasses as essential habitat. The Dwarf Seahorse (*Hippocampus zosterae*) is an understudied species in Texas, and little is known about its density, distribution, and habitat associations in this area of their range. Physicochemical water parameters, nekton community data, habitat data, and Dwarf Seahorse catch data were collected at 80 sites in Texas. The highest catch per unit effort (CPUE) of the target species was in Aransas Bay (0.038/m^2). There was a positive relationship between the presence and percent cover of turtle grass (*Thalassia testudinum*) and the presence and CPUE of Dwarf Seahorses. Dwarf Seahorses were detected more often and at a higher CPUE in locations with a higher seagrass community diversity and richness. The nekton community at sites where Dwarf Seahorses were detected was also more abundant, diverse, and species rich. This is the first comprehensive study of the distribution of the Dwarf Seahorse along the Texas coast. Dwarf Seahorses were generally found in higher abundances in association with mature, stable, and diverse seagrass beds. Recommended conservation strategy to protect Dwarf Seahorses should prioritize the protection of established and mature seagrass beds. Continued directed monitoring of this species is recommended to better understand their distribution and population status.

Keywords Seagrass · Habitat association · Macroalgae · Syngnathidae · Coastal Texas · Species status

Introduction

Seagrasses are considered foundation species because they provide critical habitat for sustaining overall species assemblages within associated ecological communities (Hughes et al. 2009). Seagrass beds provide essential nursery grounds, feeding areas, and refuge for various vertebrate and invertebrate species (Jackson et al. 2001). The health of seagrass beds is tied to recruitment and abiotic factors such as light, temperature, dissolved oxygen content, salinity, turbidity, nutrients, and disturbance levels (Hill et al. 2014). Five species of seagrass are found along the Texas Gulf Coast: shoal grass (*Halodule wrightii*, synonymous with *Halodule beaudettei*), star grass (*Halophila engelmannii*), manatee grass (*Syringodium filiforme*), turtle grass (*Thalassia testudinum*), and widgeon grass (*Ruppia maritima*).

In Texas, many marine species, including Dwarf Seahorses (*Hippocampus zosterae*), rely on seagrass beds for habitat. The Dwarf Seahorse is one of the smallest species of seahorse with an average size of 2 cm. This species resides in shallow waters throughout the Gulf of Mexico, Atlantic Coast of Florida, and Caribbean (Lourie et al. 2004). They are found primarily among seagrass, macro-algae, and coral reef habitats, which provide camouflage, anchoring structure, and protection from strong currents (Lourie et al. 2004; Wallis 2004).

Seagrasses and associated fauna face multiple environmental stressors in areas of high anthropogenic activity.

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Seagrass bed health is threatened by water pollution, direct mechanical damage, persistent trawling or dredging over and adjacent to beds, and extreme weather events (Congdon et al. 2019; Dokulil and Teubner 2010; Dolbeth et al. 2007; Kowalski et al. 2009; Longstaff and Dennison 1999; Meyer et al. 1999; Short and Wyllie-Echeverria 1996). Declines in seagrasses may be reflected quickly in associated species, as the services they provide decline at a faster rate, far before the complete disappearance of the foundation species (Ellison et al. 2005). The impacts of these threats are not well documented for Dwarf Seahorses in Texas.

Dwarf Seahorses exhibit a unique reproductive method that could influence their viability in the face of environmental threats. They are socially monogamous, maintaining the same mate at least throughout the breeding season (Masonjones and Rose 2019). Once individuals have found a mate, they participate in a four-phase courtship ritual (Masonjones and Lewis 1996). Additionally, this species exhibits genetic monogamy, meaning when partners are lost, widowed animals stop reproducing until finding a new permanent partner (Lourie et al. 2004; Rose et al. 2014). This species has a brood size smaller than most other marine fish (3–16 offspring on average) and are therefore more susceptible to population declines (Lourie et al. 2004; Masonjones and Lewis 1996). Low brood size that potentially leads to low population density means lost partners are replaced more slowly (Lourie et al. 2004).

In 2020, NOAA released a status report of the Dwarf Seahorse stating the species was not in need of protection under the Endangered Species Act (ESA) (NMFS 2020). Resource managers determined that based on historical data within the Dwarf Seahorse's range, only the east coast of Florida and northwest Florida have an elevated risk of local extirpation (NMFS 2020). They determined that this was not a substantial enough portion of their entire range to pose a threat to the overall species viability for the foreseeable future (NMFS 2020). However, also included in the report was the determination that data on the Dwarf Seahorse was sparse in all other areas of their range outside Florida. There are in-depth population estimates and confidence intervals available for many areas of the Florida coast. However, for other areas in the US Gulf Coast, Cuba, Mexico, the Bahamas, and Bermuda, there are no numerical data available, only information such as "species presence," "species absence," or "population unknown." Consistent and more wide-spread monitoring would be beneficial to inform future listing decisions. The objectives of this study were to estimate the presence and density of Dwarf Seahorses along the Texas coast and detect any correlations with physicochemical water parameters, indicators of seagrass health, and nekton community indices. Ultimately, the goal was to create a habitat suitability index reflecting the probability that a Dwarf Seahorse would be captured based on site-specific environmental variables.

Methods

Site Selection

The study area included each major estuary located between Galveston Bay and the Lower Laguna Madre, Texas (Fig. 1). Sites were sampled the summer of 2020, between June 2 and August 20. The sites selected for Galveston Bay, San Antonio Bay, Aransas Bay, Corpus Christi Bay, Upper Laguna Madre, and Lower Laguna Madre were chosen from sites historically sampled by Texas Seagrass Monitoring Program run by Dr. Ken Dunton at the University of Texas Marine Science Institute (for original site selection methodology, see Dunton et al. 2011). Texas Seagrass Monitoring Program sites that had historic Dwarf Seahorse observations nearby were selected for this project (TPWD, VertNet, Smithsonian, NOAA, iNaturalist). These sites were prioritized to represent a range of historic seagrass types and coverage and to maximize spatial coverage within the bay systems.

Matagorda and East Matagorda Bays had not been historically monitored by the Dunton Lab. These bays were divided into four sections using ArcGIS Pro, with three sections along each of the shorelines in Matagorda Bay and one in East Matagorda Bay. One site was randomly created in each of these sections where historic seagrass beds were present, based on the Texas Parks and Wildlife (TPWD) Seagrass Viewer, using the ArcGIS tool "Create Random Points" (TPWD 2016). One additional site was randomly selected along the longest coastline, the south barrier islands, for a total of five sites. The number of sites surveyed in each bay system was qualitatively based on the seagrass coverage in each bay per compiled seagrass data (TPWD 2016). Five sites were sampled each in Galveston Bay and Matagorda Bay; ten in San Antonio Bay, Aransas Bay, and Corpus Christi Bay; and twenty in Upper and Lower Laguna Madre for a total of 80 along the Texas coast (Fig. 1). The TPWD Seagrass Viewer used in site selection includes data from King and Sheridan (2006), Pulich (2007), White et al. (2002), Pulich (1991), Pulich et al. (1997), Onuf (2007), and NOAA (2007). This resource also includes layers of observational seagrass data created by TPWD staff in 2007 and 2015 that represent their best professional judgement of seagrass distribution.

Physicochemical Water Parameters

Physicochemical water parameters were collected over the course of a single summer season, once at each site before nekton and seagrass sampling. Water measurements were

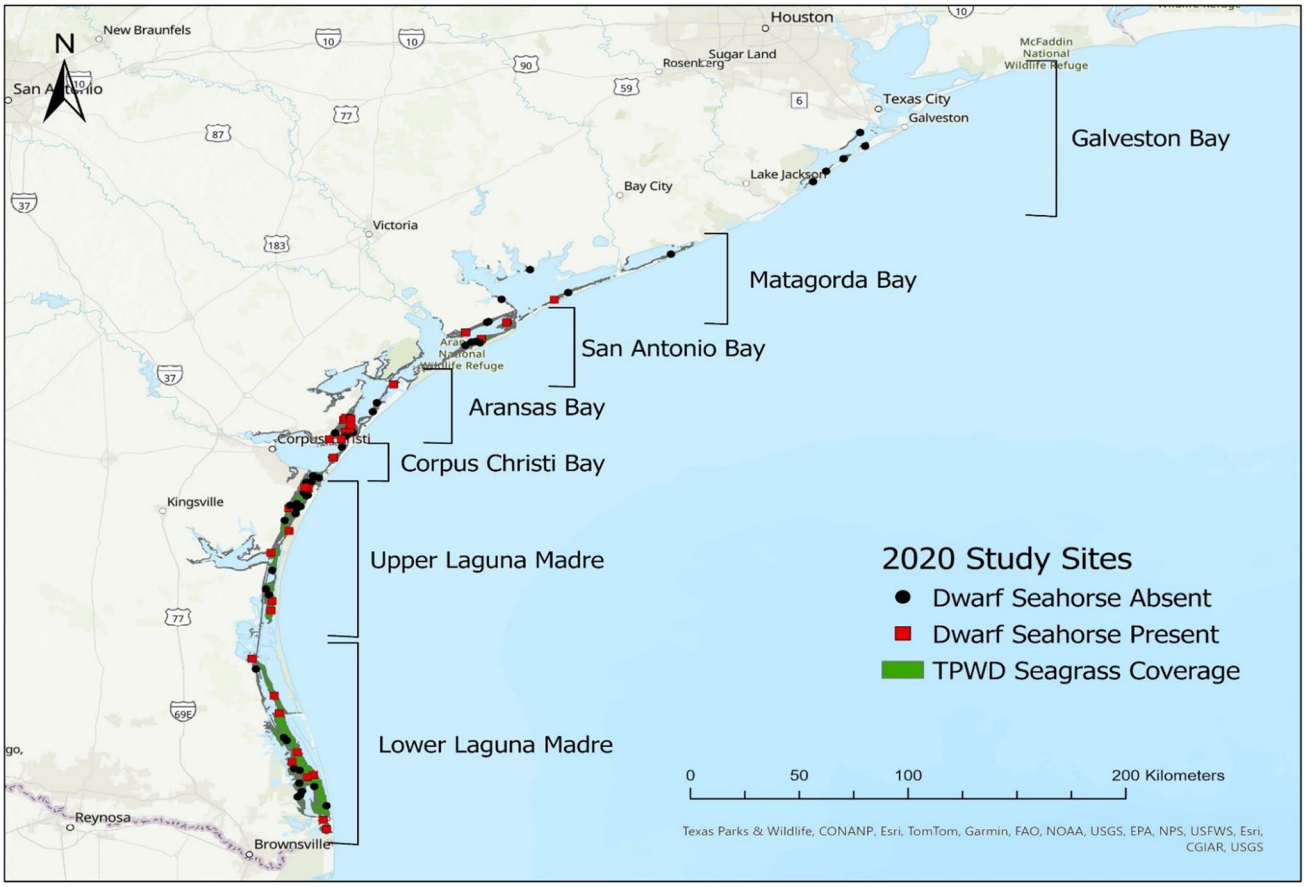

Fig. 1 Distribution of the 80 sites that were sampled for seagrass and Dwarf Seahorses in summer 2020. Black circles represent sites where Dwarf Seahorses were absent, red squares represent sites where Dwarf Seahorses were present

taken 0.1 m from the bottom at each site to record values within the seagrass bed. Using a calibrated YSI ProDSS Handheld Multiparameter Water Quality Meter, variables including depth (m), water temperature (°C), salinity (ppt), specific conductance (μMHOS/cm @ 25 °C), dissolved oxygen (mg/L), pH (standard units), and turbidity (NTU) were recorded to characterize environmental conditions within the seagrass beds. Secchi disk transparency (m) was also measured at the surface using a 1.2-m Secchi tube. Photosynthetically active radiation or PAR (μmol/$m^{-2}s^{-1}$) was measured using a LI-COR PAR meter, with an underwater reading taken at 0.1 m from the bottom of the water column and an ambient light reading taken on the boat, with the difference between these two values used for analysis.

Seagrass Bed Biomass and Percent Cover

The methodology used to measure seagrass bed community features was adapted from procedures outlined by the Global Seagrass Monitoring Network and Texas Parks and Wildlife Department (Dunton et al. 2009). Four locations within the seagrass bed were randomly selected to evaluate seagrass community composition, percent cover, biomass (g), and canopy height (cm). At each of the four random locations, a 0.25-m^2 rectangular quadrat was placed on the substrate and percent total coverage of all seagrasses and of each individual seagrass species within the quadrat were estimated, as well as any other type of bottom coverage present including bare ground, macroalgae, oysters, or debris. Average seagrass height was obtained by measuring 5 randomly selected blades, and a 15-cm diameter core was collected within the quadrat to a target depth of 20 cm below the sediment surface. The biomass sample was placed in a plastic bag, stored on ice, and taken back to lab.

Plant biomass was isolated from the replicate cores and cleaned in the laboratory. The cleaned biomass was placed in 60 °C ovens until a consistent mass was achieved to obtain dry weight, typically after 3 to 5 days. After drying, samples were placed in a desiccator until reaching room temperature. The dried biomass samples were then combined by site, weighed, and a total and average biomass weight was calculated for each site.

Seahorse and Nekton Catch

Seahorses were collected using a 1 × 1 m pushnet (0.79-mm mesh), which is pushed through the water and rolls along the bottom, collecting slow-swimming organisms in its path (Strawn 1954). The target push distance was 10 m, but if the net was filled with macroalgae and vegetation prior to the target distance, the push was stopped, and the distance of the push recorded. Up to six replicate pushes were conducted at each site. In the case of time prohibitive sampling conditions (e.g., severe weather or excessive macroalgae presence), a minimum of three pushnet replicates were completed. Any seahorses captured were euthanized using a lethal dose of MS-222 and frozen for transportation to the lab. All other nekton catch were identified and enumerated in the field. If individuals were not easily identified or catch was too numerous for field identification, all individuals were euthanized using a lethal dose of MS-222 and preserved in 10% formalin. All Dwarf Seahorses and high-quality specimens of other species kept for identification purposes were transferred to 70% ethanol for archival purposes after identification and enumeration.

Data Analysis

In this study, catch per unit effort (CPUE) refers to the number of individuals captured per m^2. For each bay system, CPUE of Dwarf Seahorses was calculated by dividing the number of individuals captured in each bay system by the total sampled area in the bay system. For the entire coast, the total number of Dwarf Seahorses captured in the study was divided by the total sampled area throughout the study. Averages of physicochemical water parameters and seagrass community metrics were calculated both by bay system and over the entire coast. Community diversity indices including seagrass species count (richness, S) as well as Pielou's evenness (J) and Shannon Diversity H (using percent cover) were calculated for the seagrass community. Calculated nekton diversity indices included number of individuals (abundance, N), relative abundance (%), and number of species present (richness, S), as well as Pielou's evenness (J) and Shannon Diversity (H) using nekton CPUE. All community diversity indices were calculated utilizing the software PAST (version 4.03). Dwarf Seahorses were included in nekton diversity calculations.

R Studio (2021.09.0) software was used to analyze the relationships between physicochemical water parameters, seagrass biomass and percent cover, seagrass and nekton community diversity, and Dwarf Seahorse presence/absence and CPUE. An $\alpha = 0.05$ was used to indicate statistical significance. Relationships between Dwarf Seahorse CPUE and physicochemical parameters, seagrass species percent cover, seagrass canopy height, seagrass biomass, seagrass community diversity indices, and nekton community diversity indices were assessed using generalized linear models and zero-inflated negative binomial regression (package pscl). For these analyses, seahorse catch was calculated per sampled area at each site (typically 60 m^2) and rounded to the nearest whole number to result in an integer. The relationship between Dwarf Seahorse CPUE and seagrass species presence was assessed using Kruskal–Wallis one-way analysis of variance. The Kruskal–Wallis one-way analysis of variance was also used to analyze relationships between Dwarf Seahorse presence/absence and physicochemical parameters, seagrass percent cover, seagrass species percent cover, seagrass canopy height, seagrass biomass, seagrass community diversity indices, and nekton diversity indices.

Statistically significant results are presented in this manuscript. For more extensive statistical results, see Oakley et al. (2022). RStudio was also used to create figures for data visualization. ArcGIS Pro was used to generate maps of spatial data. Note, when the term "absent" is used, it does not imply that Dwarf Seahorses were totally absent from that location or under those conditions, but that we were unable to detect their presence using our gear and methods.

Nekton community data were $\log(x + 1)$ transformed and a Bray–Curtis Resemblance Matrix was used to conduct non-metric multi-dimensional scaling (nMDS) using Primer 7 software. The resulting diagrams were used to compare the similarity of the nekton community observed among sites (Bray and Curtis 1957; Clarke and Warwick 2001). Differences between the nekton community at sites where Dwarf Seahorses were present and sites where Dwarf Seahorses were absent was conducted using an analysis of similarities (ANOSIM) test in Primer7. Marine species associated with the presence of Dwarf Seahorses were identified by conducting Pearson's correlation and similarity percentage (SIMPER) tests. Dwarf Seahorse catch was included in both the nekton diversity calculations and Primer community analyses.

Results

Seahorse Presence and Density

A total of 79 Dwarf Seahorses were caught at 30 of the 80 sites (Table 1). They were detected in all sampled bay systems, except Galveston Bay. Dwarf Seahorses were the only species of seahorse captured throughout this study. Dwarf Seahorse CPUE at individual sites where they were captured ranged from 0.017/m^2 to 0.136/m^2. Total CPUE across all sites was 0.017/m^2 (equivalent to catching one Dwarf Seahorse in 60 m^2 of effort). The highest bay-wide density (CPUE) of Dwarf Seahorses was detected in Aransas Bay (0.038/m^2), with lower densities found in Lower Laguna

Table 1 Summary of Dwarf Seahorse capture throughout the study by bay system. CPUE here is the total number of Dwarf Seahorses captured in each bay system divided by total sampled area (m^2)

Bay system	No. of sites	No. of Dwarf Seahorses captured	% of sites with Dwarf Seahorses	Dwarf Seahorse CPUE
Galveston	5	0	0	0.000
Matagorda	5	1	20	0.003
San Antonio	10	5	30	0.008
Aransas	10	20	60	0.038
Corpus Christi	10	6	40	0.011
Upper Laguna	20	19	35	0.017
Lower Laguna	20	28	45	0.023
Texas Coast	**80**	**79**	**37.5**	**0.017**

Madre (0.023/m^2), Upper Laguna Madre (0.017/m^2), Corpus Christi Bay (0.011/m^2), San Antonio Bay (0.008/m^2), and Matagorda Bay (0.003/m^2). There was no statistically significant difference in CPUE among bay systems ($\chi^2 = 7.7282$ (6), $p = 0.2587$, Kruskal–Wallis Rank Sum Test).

Physicochemical Water Parameters

Physicochemical parameters were collected at all 80 sites, except for PAR difference, which had a sample size of 75 due to equipment malfunction or missed readings. Since data was collected at each site once, water conditions were relatively stable and similar over the course of the project, though some trends in water quality were detected. Observationally, the upper bays (Galveston Bay, Matagorda Bay, San Antonio Bay) had higher turbidity and dissolved oxygen (Table 2). The lower bays (Aransas Bay, Corpus Christi Bay, Upper and Lower Laguna Madre) had higher water temperature and salinity (Table 2). We failed to detect any trends observationally in pH between bays (Table 2). Each physicochemical parameter was analyzed for correlation with Dwarf Seahorse presence/absence and CPUE, and no statistically significant correlations were found. Additional physicochemical water results can be found in Lesher (2022).

Seagrass

All five seagrass species known to inhabit Texas waters were detected. The highest average seagrass percent coverage was observed in the Lower Laguna Madre (63.4%), followed by the Upper Laguna Madre (57.5%), Aransas Bay (50.8%), Corpus Christi Bay (49.1%), San Antonio Bay (45.4%), Matagorda Bay (21.0%), and Galveston Bay (18.7%) (Table 3). Macroalgae was documented in all bay systems along the Texas Gulf Coast except for Galveston Bay. The highest average percentage of macroalgae coverage was recorded in Corpus Christi Bay (27.5%), with the lowest in Matagorda Bay (0.1%). Observationally, three upper bay systems had generally lower seagrass percent cover and canopy height in comparison to the four lower bay systems (Table 3). Biomass follows a similar pattern, although the Upper Laguna Madre had average biomass values more similar to the upper bays. Seagrass community structure shifts from shoal grass, widgeon grass, and star grass dominated

Table 2 Average and standard error of bottom physicochemical water parameters by bay system

Bay system	Temp (°C)	Sp. Cond (µS/cm)	Sal (ppt)	DO (mg/L)	pH	NTU	PAR Diff (µmol/m^{-2} s^{-1})	Secchi (m)	Total depth (m)
Galveston	28.30 ± 0.44	32,312 ± 2103	19.75 ± 1.31	7.77 ± 0.48	8.11 ± 0.02	13.09 ± 2.51	865.02 ± 370.83	0.23 ± 0.10	0.82 ± 0.15
Matagorda	28.56 ± 0.80	37,731 ± 971	23.85 ± 0.67	7.11 ± 0.55	8.18 ± 0.05	30.19 ± 19.09	1086.26 ± 223.58	0.38 ± 0.15	0.68 ± 0.04
San Antonio	28.44 ± 0.48	42,922 ± 1095	27.54 ± 0.80	7.64 ± 1.08	8.23 ± 0.10	12.88 ± 1.75	530.46 ± 185.65	0.37 ± 0.05	0.64 ± 0.06
Aransas	29.14 ± 0.45	49,341 ± 1328	31.93 ± 1.08	5.73 ± 0.91	8.17 ± 0.05	9.54 ± 2.46	434.26 ± 148.17	0.63 ± 0.10	0.77 ± 0.13
Corpus Christi	30.55 ± 0.79	53,245 ± 449	35.03 ± 0.36	6.01 ± 1.17	8.18 ± 0.06	3.72 ± 0.87	478.58 ± 118.06	0.93 ± 0.10	0.65 ± 0.09
Upper Laguna	30.70 ± 0.47	54,054 ± 702	35.57 ± 0.52	5.50 ± 0.66	8.12 ± 0.05	7.22 ± 1.39	564.20 ± 141.77	0.64 ± 0.08	0.90 ± 0.08
Lower Laguna	30.99 ± 0.28	39,312 ± 1992	24.99 ± 1.41	6.56 ± 0.71	8.39 ± 0.05	7.30 ± 1.53	809.09 ± 122.13	0.62 ± 0.09	0.72 ± 0.05
Overall average	**29.99 ± 0.22**	**45,908 ± 997**	**29.68 ± 0.72**	**6.37 ± 0.34**	**8.22 ± 0.03**	**9.60 ± 1.42**	**651.11 ± 64.07**	**0.59 ± 0.04**	**0.76 ± 0.03**

DO dissolved oxygen, *NTU* nephelometric turbidity unit, *PAR* photosynthetic active radiation

Table 3 Average and standard error of seagrass biomass, canopy height, and percent cover of all seagrass species, total seagrass, macroalgae, and bare ground. No biomass data is available for Galveston Bay due to data loss

Bay system	Seagrass biomass (g)	Canopy height (cm)	Shoal cover	Turtle cover	Star cover	Manatee cover	Widgeon cover	Total seagrass cover	MACRO cover	Bare cover
Galveston	N/A	3.0 ± 1.3	13.2 ± 6.0	0.0	0.3 ± 0.3	0.0	5.3 ± 5.3	18.7 ± 5.4	0.0	79.3 ± 5.3
Matagorda	0.3 ± 0.2	4.6 ± 3.1	21.0 ± 14.0	0.0	0.0	0.0	0.0	21.0 ± 14.0	0.1 ± 0.1	77.0 ± 13.4
San Antonio	0.8 ± 0.3	10.3 ± 1.4	35.0 ± 11.4	0.0	10.4 ± 5.4	0.0	0.0	45.4 ± 13.1	8.0 ± 7.2	46.6 ± 12.1
Aransas	1.5 ± 0.3	18.6 ± 3.4	9.6 ± 6.2	17.4 ± 8.0	15.9 ± 9.3	0.0	7.9 ± 5.4	50.8 ± 10.3	5.1 ± 2.1	43.9 ± 10.0
Corpus Christi	2.1 ± 1.0	20.1 ± 3.0	18.1 ± 9.0	11.9 ± 7.7	0.2 ± 0.2	18.8 ± 9.9	0.1 ± 0.1	49.1 ± 11.3	27.5 ± 7.9	23.1 ± 8.4
Upper Laguna	1.2 ± 0.2	20.9 ± 2.1	25.7 ± 8.3	0.0	20.6 ± 6.9	11.2 ± 5.0	0.0	57.5 ± 7.6	11.2 ± 4.8	28.5 ± 8.4
Lower Laguna	3.0 ± 0.7	18.2 ± 2.4	23.9 ± 7.7	30.1 ± 8.7	1.2 ± 0.9	8.2 ± 4.3	0.0	63.4 ± 7.9	6.4 ± 2.3	29.7 ± 8.1
Overall average	**1.7 ± 0.2**	**16.4 ± 1.2**	**22.4 ± 3.6**	**11.2 ± 2.9**	**8.8 ± 2.3**	**7.2 ± 2.1**	**1.3 ± 0.8**	**50.9 ± 4.0**	**9.5 ± 2.0**	**38.5 ± 4.1**

Shoal shoal grass (Halodule wrightii), *Turtle* turtle grass (Thalassia testudinum), *Star* star grass (Halophila engelmannii), *Manatee* manatee grass (Syringodium filiforme), *Widgeon* widgeon grass (Ruppia maritima), and *MACRO* macroalgae

in the upper bays to shoal grass, turtle grass, and manatee grass dominated along the lower coast.

The presence and percent cover of each seagrass species and macroalgae was analyzed for correlations with the CPUE and presence/absence of Dwarf Seahorses. Turtle grass was the only species that had a statistically significant relationship with Dwarf Seahorse presence and CPUE. The percent cover of turtle grass had a positive correlation with the presence of Dwarf Seahorses ($\chi^2 = 6.7291$, $p = 0.0095$, Kruskal–Wallis Rank Sum Test). The presence of turtle grass was also associated with Dwarf Seahorse CPUE ($\chi^2 = 8.7376$, $p = 0.0031$, Kruskal–Wallis Rank Sum Test). While not statistically significant, there is a tendency toward statistical significance between increased CPUE of Dwarf Seahorses and the presence of manatee grass ($\chi^2 = 3.6545$, $p = 0.0559$, Kruskal–Wallis Rank Sum Test), as well as Dwarf Seahorse presence/absence and canopy height ($\chi^2 = 3.1134$, $p = 0.0777$, Kruskal–Wallis Rank Sum Test). These relationships should be investigated further in future studies. Also, the higher the average biomass at a site, the more likely an individual would be captured ($\chi^2 = 4.3436$, $p = 0.0372$, Kruskal–Wallis Rank Sum Test).

Seagrass Shannon H diversity was significantly higher at sites where Dwarf Seahorses were captured in comparison to sites where they were not ($\chi^2 = 11.463$, $p = 0.0007$, Kruskal–Wallis Rank Sum Test). A statistically significant relationship was also detected between the richness of the seagrass community (seagrass species count, as a categorical variable) and CPUE of Dwarf Seahorses ($\chi^2 = 9.6569$ (4), $p = 0.0466$, Kruskal–Wallis Rank Sum Test), but post-hoc analysis showed no significant differences between seagrass species richness groups. The relationship between the number of seagrass species present and Dwarf Seahorses was further confirmed by a significant result between seagrass species richness and Dwarf Seahorse presence/absence ($\chi^2 = 9.8351$, $p = 0.0017$, Kruskal–Wallis Rank Sum Test).

Nekton Community Structure

Nekton community richness ($\chi^2 = 32.025$, $p < 0.0001$, Kruskal–Wallis Rank Sum Test) and total CPUE ($\chi^2 = 6.9265$, $p = 0.0085$, Kruskal–Wallis Rank Sum Test) had positive relationships with Dwarf Seahorse presence. Nekton community evenness had a statistically significant negative relationship with Dwarf Seahorse presence ($\chi^2 = 12.533$, $p = 0.0004$, Kruskal–Wallis Rank Sum Test) and Dwarf Seahorse CPUE ($z = 3.143$, $p = 0.0017$, zero-inflated negative binomial regression). Nekton richness ($z = 4.083$, $p < 0.0001$, zero-inflated negative binomial regression) had a positive relationship with Dwarf Seahorse CPUE.

Based on an ANOSIM analysis ($p = 0.287$), we failed to detect a statistically significant difference in nekton

community structure between sites where Dwarf Seahorses were captured and sites where they were not. Despite the lack of statistical significance, visually, there appears to be greater community similarity between sites with Dwarf Seahorse presence. Figure 2 also depicts the distribution of sites based on similarity of nekton community and six nekton species that had a Pearson's correlation value of 0.25 or greater, contributing to the data spread along the ordination 2 axis. The direction of the vector lines associated with each plotted species indicates the directional influence this species has on the data spread, while the line length indicates correlation strength (longer lines indicate a stronger correlation). Sites with Dwarf Seahorses present plot toward the bottom center area of the ordination plot. Other species that contribute to positioning a site in a similar downward direction include Gulf Toadfish (*Opsanus beta*), Rainwater Killifish (*Lucania parva*), Code Goby (*Gobiosoma robustum*), and shrimp belonging to the family Palaemonidae. This suggests that these species are found in association with Dwarf Seahorses. The results of the SIMPER analysis reflect that Palaemonidae species are most likely to co-occur with Dwarf Seahorses. In contrast, the presence of blue crabs (*Callinectes sapidus*) in an area was generally associated with lower densities of Dwarf Seahorses, as this species has a higher correlation with sites in the opposite direction on the plot. To see a complete list of captured species, see Appendix 6 in Oakley et al. (2022).

Discussion

This novel study along the Texas coast identified significant associations between Dwarf Seahorse presence/absence and the seagrass and nekton communities among which they were found. No statistically significant associations were found between measured physicochemical water parameters and Dwarf Seahorse presence/absence or density. There is little known about what water conditions Dwarf Seahorses prefer in the wild. Literature suggests that Dwarf Seahorses persist in larger numbers in areas of lower turbidity (Erftemeijer and Lewis 2006). This aligns with our observations, where we detected an increase in Dwarf Seahorse CPUE in lower bay systems with lower average turbidity levels.

Each site was only sampled once for this study and all sampling took place throughout one summer season, meaning water conditions were relatively consistent. The physicochemical data we collected was a snapshot of the conditions during sampling and not representative of long-term trends in water quality throughout the study area. This fact makes it difficult to detect large-scale patterns in physiochemical variables that may be influential in spatial distribution of

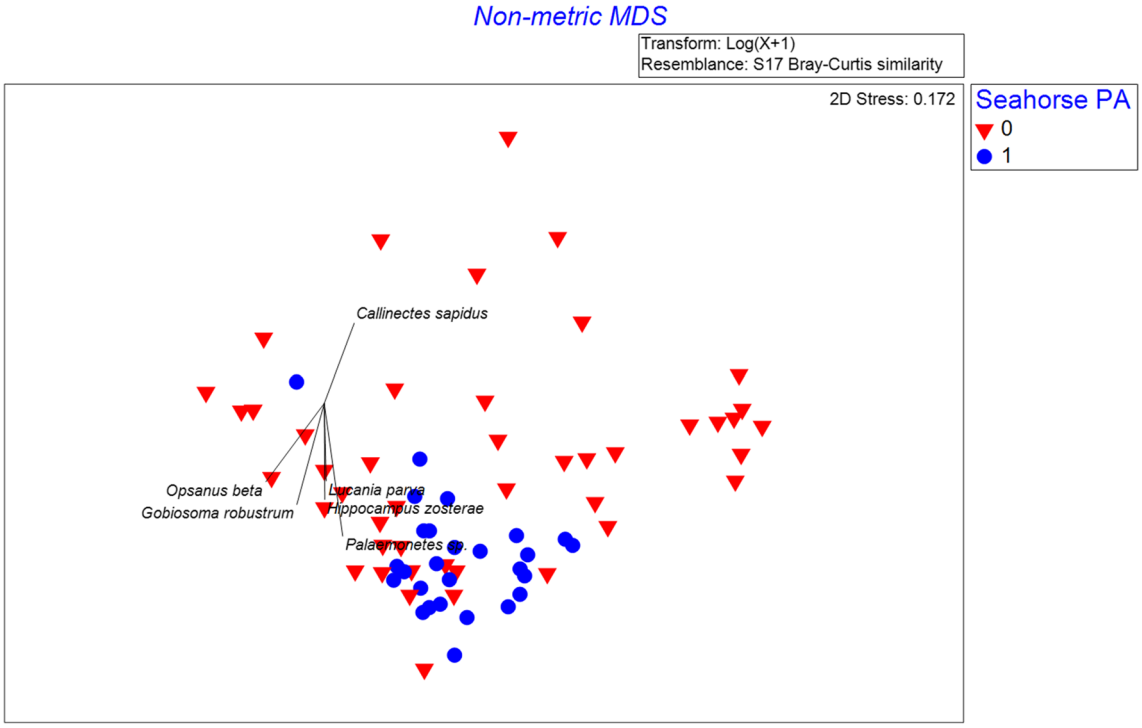

Fig. 2 Non-metric multidimensional scaling plot generated by Primer 7 of log(x + 1) transformed nekton community. Blue circles represent sites where Dwarf Seahorses were observed and red triangles represent sites where they were not observed. Species listed on figure have Pearson's correlation of over 0.25 on the ordination 2 axis (y axis)

seagrasses and associated low mobility species such as the Dwarf Seahorse. However, typical spatial trends in water quality were observed relative to a temperature, precipitation, and freshwater inflow gradient along the coast, affecting water salinity and turbidity (Kinard et al. 2021). The northeastern portion of the Texas coast experiences more freshwater inflows, higher annual precipitation, and ranks higher on the Global Aridity Index, meaning it is less arid (Osland et al. 2014). Moving southwest, freshwater inflows and annual precipitation decrease while aridity increases (Osland et al. 2014). Our data for water temperature, salinity, specific conductivity, and turbidity align with what was expected based on this gradient and Dwarf Seahorses seem to prefer the conditions found along the lower coast. A significant relationship between Dwarf Seahorse catch and physicochemical water parameters may be undetected in this study due to low target species catch and sites being sampled a single time throughout the summer, not reflecting long-term trends and relationships. Additionally, these spatial observations in water conditions likely contribute to the increased seagrass coverage seen along the lower coast, providing more habitat for Dwarf Seahorses to reside.

These consistent conditions were disturbed during some of our sampling efforts. Some sites in the Upper Laguna Madre were sampled just 3 days after category 1 Hurricane Hanna hit the Padre Island area (Brown et al. 2021). The storm directly affected the Texas coast July 24–26, 2020, and eight Upper Laguna Madre sites were sampled the week following. At this time, the area was still experiencing higher water levels than usual, which potentially influenced Dwarf Seahorse catch, vegetation analysis, and physicochemical parameters. Increased water depth may have influenced the efficacy of pushnet sampling. The high winds and storm surge also could have displaced vegetation (both detached macroalgae and floating mats of seagrass) that contained Dwarf Seahorses. Alternatively, the storm also could have moved individuals into our sampling area that are not typically residents. Dwarf Seahorse CPUE in this area could have been out of the ordinary, but no sites in the Upper Laguna Madre were sampled prior to the storm and there is no consistent historic data to compare to. We acknowledge that the Dwarf Seahorse catch at the eight sites sampled immediately after Hurricane Hanna may not be representative of typical densities during baseline conditions.

Even with this disturbance, results throughout this study reinforce existing knowledge of the associations between Dwarf Seahorses and seagrass. We primarily found Dwarf Seahorses residing in dense seagrass beds with higher canopy heights (Lourie et al. 2004; Masonjones et al. 2010). This project detected significant relationships between increased canopy height and biomass and Dwarf Seahorse presence. We saw significant relationships between Dwarf Seahorses and turtle grass, the seagrass species with the highest biomass in Texas. Both the presence and percent cover of this seagrass species had significantly positive correlation with Dwarf Seahorse CPUE and presence, though Dwarf Seahorses were also found in beds with all other Texas seagrass species.

There are a few potential explanations for this observed relationship. First, as mentioned, turtle grass has the highest biomass of all Texas seagrass species and a tall canopy height. These features may help slow water velocity and currents in areas where the species densely populates, protecting slow moving organisms such as Dwarf Seahorses (Fonseca et al. 1982). Secondly, turtle grass is a climax species; its presence indicates the ecosystem is stable, with optimal water conditions (saline and low turbidity levels) and has been established in the area for substantial time (Congdon et al. 2019). Relatedly, results from this study also indicate a potential relationship between the seagrass biodiversity observed at a site and Dwarf Seahorse presence and CPUE. Shannon H diversity of the seagrass community was positively correlated with Dwarf Seahorse presence. A diverse seagrass bed indicates a mature, established bed, which would be preferred by this small, slow-moving species.

The association between manatee grass presence and Dwarf Seahorse CPUE was not statistically significant but indicated a potential positive relationship. The heartiness of manatee and turtle grasses and their relationships with Dwarf Seahorse presence and CPUE likely explain the similar positive associations with seagrass average biomass and canopy height, or vice versa. This finding could potentially indicate that Dwarf Seahorses prefer larger seagrass species that have the potential to provide more protection and stable habitat conditions. However, during this study, a Dwarf Seahorse was captured at a site with only macroalgae cover observed. While diverse and mature seagrass beds seem to support more consistent and robust Dwarf Seahorse numbers in other parts of their range, it is possible that Dwarf Seahorses are able to utilize other habitats where seagrasses are not present.

It is also important to note that throughout sampling, we observed that the four randomly distributed quadrats used for the seagrass quantification data did not always summarize all seagrass species present in the area. While sampling, we observed seagrass species present within our sampling area that were not included within our four quadrats. Further study of the associations between Dwarf Seahorses and seagrass is needed to describe these relationships more confidently. Considering ways to better document seagrass community at the precise locations where Dwarf Seahorses are observed should be considered when planning future studies.

The relationship between seagrass beds and residing Dwarf Seahorses may differ as environmental conditions change along the Texas coast. Environmental variables such as turbidity, salinity, and human activity levels influence the

success and recovery of seagrass beds (Chislock et al. 2013; Dolbeth et al. 2007; Hill et al. 2014; Longstaff and Dennison 1999; Meyer et al. 1999; Short and Wyllie-Echeverria 1996). There are known salinity and turbidity gradients along the Texas coast (Bugica et al. 2020; Kinard et al. 2021). As mentioned earlier, the upper bays have higher turbidity levels and lower salinity due to higher precipitation levels and freshwater inflows (Bugica et al. 2020; Osland et al. 2014). Other activities like channelization, shoreline development, and wastewater and pollutant discharge can contribute to increased turbidity and eutrophication as well (Pulich and White 1991). These combined factors have contributed to losses of seagrass in some areas of the upper Texas bay systems (Pulich and White 1991). The lower bays tend to have less coastal human influence and freshwater inflow, creating a better environment for seagrass proliferation. We saw the implications of habitat quality degradation reflected in our Dwarf Seahorse catch. Upper coast bays had less turtle grass and lower seagrass species presence and percent cover, so Dwarf Seahorses lacked diverse seagrass bed habitat similar to that found in lower bay areas, where there is higher species richness and percent coverage.

The diversity of the residing nekton community also had positive correlation with Dwarf Seahorse presence throughout coastal Texas. Nekton community evenness, richness, and abundance in CPUE were all positively correlated with Dwarf Seahorse presence, and community evenness and richness were positively correlated with Dwarf Seahorse CPUE. The nekton community assemblages at sites examined throughout this study were variable. At some sites, only ctenophores and other invertebrates were detected while at other sites, there were hundreds of individuals from multiple species captured; a few sites had no catch at all. There are many benefits to a diverse community assemblage. A robust abundance and variety of individuals from all trophic levels in an ecosystem provide a necessary balance and resilience to maintain community structure (Sala and Knowlton 2006). Areas of lower diversity are more susceptible to stress during disturbance (Sala and Knowlton 2006). Dwarf Seahorses seem to be more commonly observed as part of a diverse nekton community which resides in mature and diverse seagrass habitats.

While it is known that high nekton diversity is beneficial for community stability, less studied are the impacts that "rare" species have on their communities. Dwarf Seahorses could be considered "rare" based on its small known population, small brood size, limited geographic range, and potentially narrow habitat tolerance. Because of this, the presence of Dwarf Seahorses may indicate hotspots of biodiversity along the Texas coast. The presence of rare species positively impacts functional richness, specialization, and originality of assemblages, as reflected in random loss of species scenarios (Leitão et al. 2016). Leitão et al. (2016) demonstrate that not only dominant species play a critical role, but rare species are also functionally important in maintaining ecological processes. This phenomenon may partially explain why the presence of this species was correlated with multiple diversity metrics.

In terms of associated species, more research is needed to determine relationships and dynamics between Gulf Toadfish, Rainwater Killifish, Code Goby, blue crab, Palaemonidae shrimp species, and Dwarf Seahorse. Though this relationship is not documented for this species, it is possible that small larval shrimp serve as a source of prey for Dwarf Seahorses. This relationship could explain why shrimp are found in high numbers where Dwarf Seahorses are present. Conversely, blue crabs are predators of this species, potentially explaining why Dwarf Seahorse presence decreased when blue crab were present in high numbers (Strawn 1958). Gulf Toadfish, Rainwater Killifish, and Code Gobies may benefit from inhabiting these same stable, mature seagrass beds. Further study may discern other explanations for associations between these species.

In summary, this is the first comprehensive study of the distribution of the Dwarf Seahorse along the Texas coast. Furthermore, we have identified a few significant factors that correlate with the presence/absence and density of Dwarf Seahorses along the Texas coast. As mentioned in the introduction, the goal of this study was to ultimately create a habitat suitability model for Dwarf Seahorses along the Texas coast. Fewer variables were found to significantly influence Dwarf Seahorse presence/absence and CPUE than expected, making it difficult to create a comprehensive, accurate habitat suitability model. More extensive, in-depth future studies could work toward this goal with more success.

Sampling for this species is a challenging endeavor. The use of more physically demanding gear types is necessary, as traditional gear types used by state fisheries monitoring agencies have proved to be ineffective in catching this species (Oakley et al. 2022). Additionally, hand picking through vegetation to identify this small and somewhat elusive species is incredibly time consuming. Despite these challenges, it is important to understand baseline density, habitat use, and distribution of this species in Texas to track changes over time and inform management decisions. Locations in the mid to lower Texas coast with established seagrass beds and nekton communities including Gulf Toadfish, Rainwater Killifish, Code Goby, and Penaeid shrimp should be prioritized for future study and conservation. Future work could expand on these findings to create a habitat suitability index estimating the probability of finding a Dwarf Seahorse in certain locations. More in-depth and comprehensive research is needed on this understudied species in all areas of their range. It is only through long-term continued monitoring,

using effective gear, and standardizing methods that trends can be evaluated, and Dwarf Seahorse population viability status be determined.

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Data Availability Data is not available on any online platform, but available by request.

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References

Bray, J.R., and J.T. Curtis. 1957. An ordination of upland forest communities of southern Wisconsin. *Ecological Monographs* 27: 325–349.

Brown, D. P., R. Berg, and B. Reinhart. 2021. Hurricane hanna. National hurricane center tropical cyclone report. NOAA and NWS. https://www.nhc.noaa.gov/data/tcr/AL082020_Hanna.pdf. Accessed 1 Apr 2022.

Bugica, K., B. Sterba-Boatwright, and M.S. Wetz. 2020. Water quality trends in Texas estuaries. *Marine Pollution Bulletin* 152: 110903.

Chislock, M.F., E. Doster, R.A. Zitomer, and A.E. Wilson. 2013. Eutrophication: Causes, consequences, and controls in aquatic ecosystems. *Nature Education Knowledge* 4 (4): 10.

Clarke, K. R., and R. M. Warwick. 2001. *Change in marine communities: an approach to statistical analysis and interpretation.* 2nd edition. Plymouth: PRIMER-E Ltd.

Congdon, V.M., C. Bonsell, M.R. Cuddy, and K.H. Dunton. 2019. In the wake of a major hurricane: differential effects on early vs. late successional seagrass species. *Limnology and Oceanography Letters* 4(5): 155–163.

Dokulil, M. T., and K. Teubner. 2010. Eutrophication and climate change: present situation and future scenarios. In *Eutrophication: causes, consequences, and control*, ed. Abid A. Ansari and Sarvajeet Singh Gill, 1–16. Dordrecht: Springer.

Dolbeth, M., P.G. Cardoso, S.M. Ferreira, T. Verdelhos, D. Raffaelli, and M.A. Pardal. 2007. Anthropogenic and natural disturbance effects on a macrobenthic estuarine community over a 10-year period. *Marine Pollution Bulletin* 54 (5): 576–585.

Dunton, K., K. Jackson, and C. Wilson. 2009. Seagrass and macroalgal biomass in tier 2 seagrass protocols. Texas Seagrass Monitoring Program. https://utw10133.utweb.utexas.edu/doc/Seagrass_Protocols.pdf. Accessed 15 Jan 2020.

Dunton, K. H., W. Pulich, and T. Mutchler. 2011. A seagrass monitoring program for Texas coastal waters: multiscale integration of landscape features with plant and water quality indicators. Texas Seagrass Monitoring Program. https://dev-texasseagrass.pantheonsite.io/wp-content/uploads/2023/06/A-Seagrass-Monitoring-Program-for-Texas-1-10-11.pdf. Accessed 15 Jan 2020.

Ellison, A.M., M.S. Bank, B.D. Clinton, E.A. Colburn, K. Elliott, C.R. Ford, D.R. Foster, B.D. Kloeppel, J.D. Knoepp, G.M. Lovett, J. Mohan, D.A. Orwig, N.L. Rodenhouse, W.V. Sobczak, K.A. Stinson, J.K. Stone, C.M. Swan, J. Thompson, B. Von Holle, and J.R. Webster. 2005. Loss of foundation species: Consequences for the structure and dynamics of forested ecosystems. *Frontiers in Ecology and the Environment* 3 (9): 479–486.

Erftemeijer, P.L., and R.R.R. Lewis III. 2006. Environmental impacts of dredging on seagrasses: A review. *Marine Pollution Bulletin* 52 (12): 1553–1572.

Fonseca, M.S., J.S. Fisher, J.C. Zieman, and G.W. Thayer. 1982. Influence of the seagrass, *Zostera marina* L., on current flow. *Estuarine, Coastal and Shelf Science* 15(4): 351–364.

Hill, V.J., R.C. Zimmerman, W.P. Bissett, H. Dierssen, and D.D. Kohler. 2014. Evaluating light availability, seagrass biomass, and productivity using hyperspectral airborne remote sensing in Saint Joseph's Bay, Florida. *Estuaries and Coasts* 37 (6): 1467–1489.

Hughes, A.R., S.L. Williams, C.M. Duarte, K.L. Heck Jr., and M. Waycott. 2009. Associations of concern: Declining seagrasses and threatened dependent species. *Frontiers in Ecology and the Environment* 7 (5): 242–246.

Jackson, J.B., M.X. Kirby, W.H. Berger, K.A. Bjorndal, L.W. Botsford, B.J. Bourque, R.H. Bradbury, R. Cooke, J. Erlandson, J.A. Estes, T.P. Hughes, S. Kidwell, C.B. Lange, H.S. Lenihan, J.M. Pandolfi, C.H. Peterson, R.S. Steneck, M.J. Tegner, and R.R. Warner. 2001. Historical overfishing and the recent collapse of coastal ecosystems. *Science* 293 (5530): 629–637.

Kinard, S., C.J. Patrick, and F. Carvallo. 2021. Effects of a natural precipitation gradient on fish and macroinvertebrate assemblages in coastal streams. *PeerJ* 9: e12137.

King, S.P., and P. Sheridan. 2006. Nekton of new seagrass habitats colonizing a subsided salt marsh in Galveston Bay. *Estuaries* 29 (2): 286–296.

Kowalski, J.L., H.R. DeYoe, and T.C. Allison. 2009. Seasonal production and biomass of the seagrass, *Halodule wrightii* Aschers. (shoal grass), in a subtropical Texas lagoon. *Estuaries and Coasts* 32(3): 467–482.

Leitão, R.P., J. Zuanon, S. Villéger, S.E. Williams, C. Baraloto, C. Fortunel, F.P. Mendonça, and D. Mouillot. 2016. Rare species contribute disproportionately to the functional structure of species assemblages. *Proceedings of the Royal Society B: Biological Sciences* 283 (1828): 20160084.

Lesher, S.C. 2022. Habitat use, distribution, and density of dwarf seahorse (*Hippocampus Zosterae*) populations along the Texas Gulf Coast. Master's Thesis. University of Houston – Clear Lake, Houston. 106pp.

Longstaff, B.J., and W.C. Dennison. 1999. Seagrass survival during pulsed turbidity events: The effects of light deprivation on the

seagrasses *Halodule pinifolia* and *Halophila ovalis*. *Aquatic Botany* 65 (1–4): 105–121.

Lourie, S. A., S. J. Foster, E. W. Cooper, and A. C. Vincent. 2004. A guide to the identification of seahorses. CITES Technical Manual. https://projectseahorse.org/resource/a-guide-to-the-identification-of-seahorses/. Accessed 15 Jan 2020.

Masonjones, H.D., and S. Lewis. 1996. Courtship behavior in the dwarf seahorse, *Hippocampus zosterae*. *Copeia* 1996: 634–640.

Masonjones, H.D., E. Rose, L.B. Mcrae, and D.L. Dixson. 2010. An examination of the population dynamics of syngnathid fishes within Tampa Bay, Florida, USA. *Current Zoology* 56 (1): 118–133.

Masonjones, H.D., and E. Rose. 2019. When more is not merrier: Using wild population dynamics to understand the effect of density on ex situ seahorse mating behaviors. *PLoS ONE* 14 (7): e0218069.

Meyer, D.L., M.S. Fonseca, P.L. Murphey, R.H. McMichael, M.W. LaCroix, P.E. Whitfield, and G.W. Thayer. 1999. Effects of live-bait shrimp trawling on seagrass beds and fish bycatch in Tampa Bay, Florida. *Fishery Bulletin* 97 (1): 193–199.

National Marine Fisheries Service (NMFS). 2020. Status review report: dwarf seahorse (*Hippocampus zosterae*). National Marine Fisheries Service, Office of Protected Resources. https://www.fisheries.noaa.gov/s3/dam-migration/2017-dwarf-seahorse-status-review-final-508-1-.pdf. Accessed 1 Apr 2022.

National Oceanic and Atmospheric Administration (NOAA). 2007. Coastal Bend Texas Benthic habitat mapping patchy shapefile map. *NOAA Office for Coastal Management Metadata Library*. https://www.fisheries.noaa.gov/inport/item/47960. Accessed 1 March 2020.

Oakley, J. W., S. Lesher, G. Dennis, R. Ulibarri, and G. Guillen. 2022. Distribution, abundance, and habitat use of the dwarf seahorse, *Hippocampus zosterae*, along the Texas coast. Environmental Institute of Houston. https://www.uhcl.edu/environmental-institute/research/publications/documents/21-001-dwarf-seahorse-final-report.pdf. Accessed 1 Dec 2021.

Onuf, C.P. 2007. Laguna Madre. In *Seagrass status and trends in the Northern Gulf of Mexico: 1940–2002*, eds. L. Handley, D. Altsman, R. DeMay, 29–40. U.S. Geological Survey Scientific Investigations Report 2006–5287 and U.S. EPA 855-R-04–003.

Osland, M.J., N. Enwright, and C.L. Stagg. 2014. Freshwater availability and coastal wetland foundation species: Ecological transitions along a rainfall gradient. *Ecology* 95 (10): 2789–2802.

Pulich, W., Jr. 1991. Effects of freshwater inflows on estuarine vegetation of the San Antonio Bay system. Special Project Report. Submitted to Austin, Texas Parks and Wildlife Department, Resource Protection Division. 31 pp., 12 figs.

Pulich, W.M., Jr., and W.A. White. 1991. Decline of submerged vegetation in the Galveston Bay system: Chronology and relationship to physical processes. *Journal of Coastal Research* 7: 1125–1138.

Pulich, W., C. Blair, and W. A. White. 1997. Current status and historical trends of seagrasses in the Corpus Christi Bay National Estuary program study area. TPWD Coastal Studies Program. https://www.cbbep.org/publications/virtuallibrary/CC20.pdf. Accessed 15 Jan 2022.

Pulich, W. 2007. Galveston Bay system. In *Seagrass status and trends in the Northern Gulf of Mexico: 1940–2002*, eds. L. Handley, D. Altsman, R. DeMay, 17–28. U.S. Geological Survey Scientific Investigations Report 2006–5287 and U.S. EPA 855-R-04–003.

Rose, E., C.M. Small, H.A. Saucedo, C. Harper, and A.G. Jones. 2014. Genetic evidence for monogamy in the dwarf seahorse, *Hippocampus zosterae*. *Journal of Heredity* 105 (6): 828–833.

Sala, E., and N. Knowlton. 2006. Global marine biodiversity trends. *Annual Review of Environment and Resources* 31: 93–122.

Short, F.T., and S. Wyllie-Echeverria. 1996. Natural and human-induced disturbance of seagrasses. *Environmental Conservation* 23 (1): 17–27.

Strawn, K. 1954. The pushnet, a one-man net for collecting in attached vegetation. *Copeia* 1954 (3): 195–197.

Strawn, K. 1958. Life history of the pigmy seahorse, Hippocampus zosterae Jordan and Gilbert, at Cedar Key, Florida. *Copeia* 1958 (1): 16–22.

Texas Parks and Wildlife. 2016. TPWD Seagrass viewer. ArcGIS Online. https://tpwd.maps.arcgis.com/apps/webappviewer/index.html?id=af7ff35381144b97b38fe553f2e7b562. Accessed 1 Mar 2020.

Wallis, C. 2004. *Seahorses: Mysteries of the oceans*. Chicago: Bunker Hill Publishing Inc.

White, W. A., T. R. Tremblay, R. L. Waldinger, and T. R. Calnan. 2002. Status and trends of wetland and aquatic habitats on Texas Barrier Islands, Matagorda Bay to San Antonio Bay. Texas GLO Coastal Coordination Division. https://www.cbbep.org/publications/virtuallibrary/BarrierWetlandsCoastalBend.pdf. Accessed15 Jan 2022.